

Spatial and Geographic Impacts of Climate Change on Scottish Whisky Terroir: A Micro-Level Spatial Econometric and Clustering Analysis

Donghyun Kim, Donghyuk Lee, Yeonjae Park
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github.com/donghyuklee1/HSS39943-whisky-analysis

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Abstract

This study utilizes Geographic Information Systems (GIS) and localized spatial statistics to examine the highly uneven geographic distribution of climate change exposure across the Scottish whisky industry. Rather than relying on aggregated regional boundaries, we model 226 active distilleries extracted from Wikidata as discrete point features. Future climate projections for the year 2050 are acquired from the CMIP6 framework (MPI-ESM1-2-XR model) via the Open-Meteo API and interpolated into continuous surfaces using Inverse-Distance Weighting (IDW). To capture localized, non-stationary environmental dynamics, we employ Geographically Weighted Regression (GWR) alongside K-means clustering to delineate risk tiers. Environmental exposure is quantified via a multi-criteria Composite Vulnerability Index (CVI) integrating projected shifts in annual mean temperature, precipitation, and relative humidity. Exposure is strongly uneven in space: island and northern coastal margins, together with parts of the southern Lowlands, rank high, while the north-eastern Speyside basin ranks low. This index measures projected exposure rather than realised production impacts, and some results depend on interpolation behaviour at data boundaries and on how the index itself is constructed. With these limitations stated up front, the study extends prior GIS work by moving to point-level resolution, local (non-stationary) modelling, and a 2050 projection horizon.

1 Introduction

The structural identity, sensory complexity, and historical branding of Scotch whisky are deeply intertwined with the environmental matrices of the geographic locales where production takes place. The unique regional characteristics of whisky are shaped not only by localized distillation techniques but also by the explicit macro- and micro-climatic conditions, hydrologic purity, and macroeconomic landscape of the production site [1, 2]. This deterministic relationship between a geographic place and the sensory output of an agricultural or spirituous product is captured by the concept of *terroir*, an analytical framework originally matured within viticultural science.

As formulated by van Leeuwen and Seguin (2006), terroir represents an interactive ecosystem wherein regional climate, geomorphology, soil composition, and plant genetics engage in a multi-layered feedback loop that dictates final product quality, style, and organoleptic typology [2]. Within viticulture, this mechanism operates directly via the biological status of the vine, particularly root-zone water availability. For distilled spirits like whisky, however, the translation of terroir requires a nuanced adaptation. Because malting barley is frequently sourced via globalized or non-local supply chains, much of a distillery’s terroir is physically localized during mashing, fermentation, water abstraction, and long-term barrel maturation within warehouse micro-environments [1]. Thus, rather than directly transplanting viticultural mechanisms, this study constructs a conservative geographic framing: evaluating how localized climate change exposures disrupt the fundamental physical pillars of whisky production—namely water availability, local barley yields, and cask maturation stability.

The extreme environmental dependence of the Scotch whisky industry leaves its spatial infrastructure uniquely vulnerable to systemic climate disruptions. Empirical climate data published by Adaptation Scotland (2025) demonstrate that Scotland’s ten warmest years on record have all manifested post-1997. Furthermore, the decadal window from 2014 to 2023 experienced an average temperature elevation of 1.02°C relative to the historical 1961–1990 baseline, accompanied by a 10% expansion in annual precipitation volume, disproportionately weighted toward winter surges (+29%) [1]. Climatic projections indicate a progressive shift toward hotter, hyper-arid summers and hyper-humid, warmer winters. Crucially, the James Hutton Institute (CREW) projects that severe water-scarcity events affecting industrial operations could double in frequency by the year 2050 [1]. Such shifts threaten to suppress localized barley yields, induce microbial instability during fermentation, and structurally alter warehouse relative humidity, thereby altering the historical evaporation dynamics within aging casks.

Despite these mounting risks, existing geographic literature remains highly aggregated. Prior spatial analysis, most notably exemplified by LaMunyon (2025), integrated historical Met Office grids and governmental crop statistics to model climate impacts on barley yields [3]. While LaMunyon established a weak global negative correlation between historical temperature and barley yields ($r = -0.21$) and a weak positive correlation with rainfall ($r = +0.14$), the analysis operated exclusively at the broad administrative-district level [3]. Consequently, it relied entirely on global stationary statistics, leaving the localized, point-level variations of individual distilleries unresolved, and lacked predictive mapping for future horizons.

To address these empirical and methodological gaps, this study presents a high-resolution, point-level spatial assessment of climate change exposure across the Scottish whisky sector. The core objectives are four-fold: (1) to map and analyze 226 active distilleries as individual point vectors; (2) to model regional spatial non-stationarity via Geographically Weighted Regression (GWR); (3) to project high-resolution exposure surfaces for a 2050 horizon utilizing the CMIP6 ensemble framework; and (4) to synthesize these multidimensional stressors into an interpretation-driven Composite Vulnerability Index (CVI).

2 Data & Methodology

2.1 Data Sources and Pre-processing

This research integrates four primary open-source datasets to model spatial infrastructure and environmental vectors. To ensure spatial accuracy and geometric alignment across all localized analysis steps, all spatial features were unified to the British National Grid projected coordinate system (EPSG:27700).

Table 1: Systematic Overview of the Multidimensional Open Datasets

Data Category	Primary Source	Operational Role and Analytical Function
Distillery Infrastructure	Wikidata (SPARQL)	Extraction of coordinates and nomenclatures for 226 distilleries; bounded by Lat 54.5–61.0, Lon -8.5– -0.5; operational categorization based on functional type.
Climate Projections	Open-Meteo API	CMIP6 framework (MPI-ESM1-2-XR); delta metrics to 2050 for temperature (ΔT), precipitation (ΔP), and relative humidity (ΔRH).
Geographic Polygon	GADM v4.1	Extraction and masking of Scotland Level-1 administrative data from the United Kingdom master boundary.
Cartographic Basemap	CartoDB Positron	High-legibility web tiles deployed via Contextily for background visualization.

The primary infrastructural coordinates were extracted utilizing a customized SPARQL query targeting active distillation facilities within Scotland. The geographic distributions and functional activity types—classified by whether a site conducts maturation exclusively, production exclusively, or integrated production and maturation—are presented below.

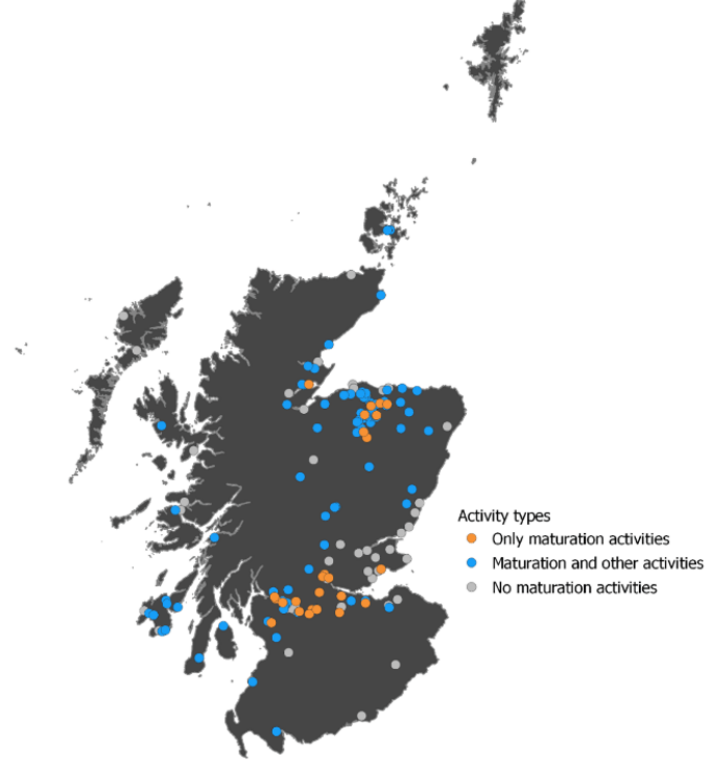


Figure 1: Spatial distribution and operational functional classification of the 226 analyzed Scottish whisky distilleries.

The macroclimatic basis utilizes a single CMIP6 climate model (MPI-ESM1-2-XR). Given this reliance on a single scenario model, absolute figures are interpreted as point estimates representing specific trajectory tendencies, as comprehensive multi-model ensemble uncertainty quantification falls outside the scope of this study.

2.2 Analytical Workflow and Spatial Formulation

The methodology executes a five-stage analytical workflow: dataset harmonization, risk tier validation, localized spatial econometric modeling, continuous surface interpolation, and internal structural diagnostics.

To divide the 226 distilleries into objective risk tiers, K-means clustering was applied over the spatial coordinates and baseline climate variables. The mathematical minimization of the within-cluster sum of squares ($WCSS$) is structured as:

$$WCSS = \sum_{j=1}^k \sum_{i \in C_j} \|x_i - \mu_j\|^2 \quad (1)$$

where C_j represents the j -th cluster, x_i is the multidimensional feature vector of distillery i , and μ_j is the centroid of cluster C_j . The optimal number of clusters k was determined utilizing a joint evaluation of the Elbow Method and Silhouette Width analysis.

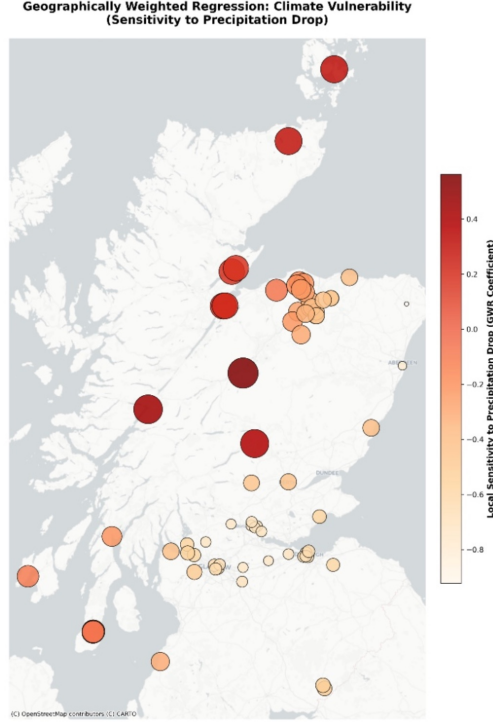


Figure 2: Statistical validation plots for K-means parameter optimization: Elbow Method (left) and Silhouette Analysis (right).

As demonstrated in Figure 2, the $WCSS$ curve exhibits a distinct bend at $k = 3$. Although global silhouette widths peak at $k = 2$, this study intentionally enforces a local choice of $k = 3$ within a restricted operational range to facilitate a highly interpretable three-tier classification corresponding to 'Safe', 'Warning', and 'Critical' environmental risk profiles.

To address spatial non-stationarity—wherein the statistical relationships between geographic variables vary across space—we depart from standard global Ordinary Least Squares (OLS) models. We deploy Geographically Weighted Regression (GWR), which fits localized regression coefficients by positioning a spatial kernel over each unique distillery coordinate:

$$Y_i = \beta_0(u_i, v_i) + \beta_1(u_i, v_i)X_{1i} + \epsilon_i \quad (2)$$

where (u_i, v_i) denotes the projected coordinate pair of the i -th distillery point, Y_i is the dependent exposure metric, X_{1i} denotes the localized precipitation decline vector, $\beta_0(u_i, v_i)$ is the local intercept, and $\beta_1(u_i, v_i)$ represents the localized parameter estimate mapping the spatial sensitivity to precipitation drop.

To transform point-level climate observations into continuous geographic layers, Inverse-Distance Weighting (IDW) interpolation was executed across the Scottish landmass. These continuous grids were subsequently compiled into a normalized 0 to 100 Composite Vulnerability Index (CVI), calculated as:

$$CVI = 0.40 \cdot \Delta P + 0.30 \cdot \Delta T + 0.20 \cdot \Delta RH \quad (3)$$

where ΔP is the percentage reduction in precipitation, ΔT is the mean temperature rise, and ΔRH represents the change in relative humidity. The assigned coefficients represent nominal design weights reflecting *a priori* empirical assumptions regarding water access and storage infrastructure.

3 Results

3.1 Localized Sensitivity and Baseline Econometric Modeling

The localized parameter surfaces derived via GWR illuminate extensive non-stationarity in the precipitation sensitivity mapping.

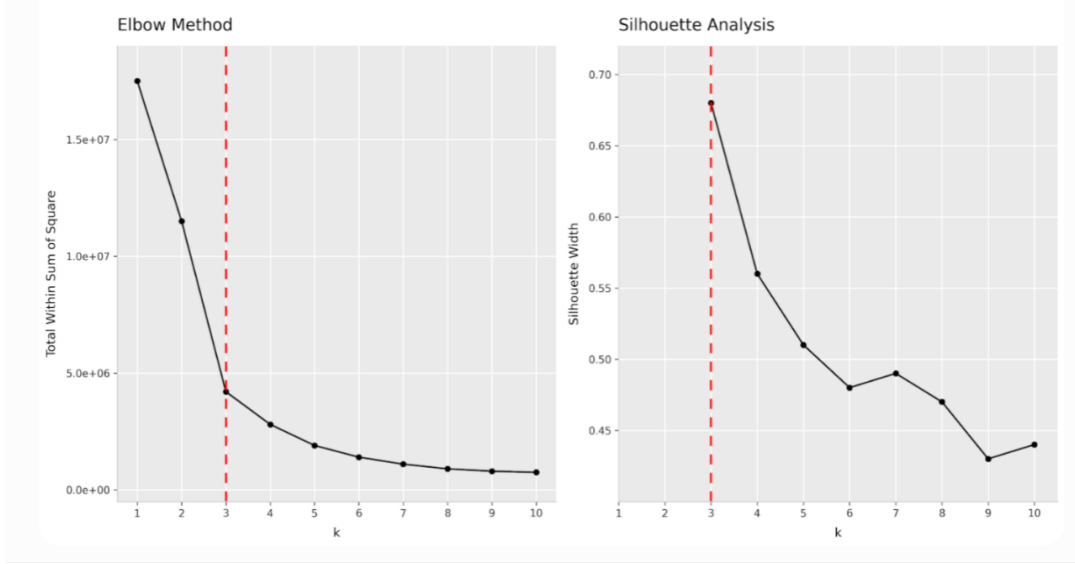


Figure 3: Geographically Weighted Regression (GWR) local coefficient (β_1) surface mapping distillery sensitivity to precipitation drops.

As illustrated in Figure 3, the highest positive localized regression coefficients ($\beta_1 \geq 0.4$) are heavily concentrated along the northern Highlands and the far-northern Orkney margin. Conversely, regions currently experiencing large historical precipitation volatility, such as Islay in the south-west, exhibit lower local parameter variance. This statistical behavior flags the northern administrative margins as a critical geographic tipping zone highly sensitive to marginal shifts in future precipitation decline.

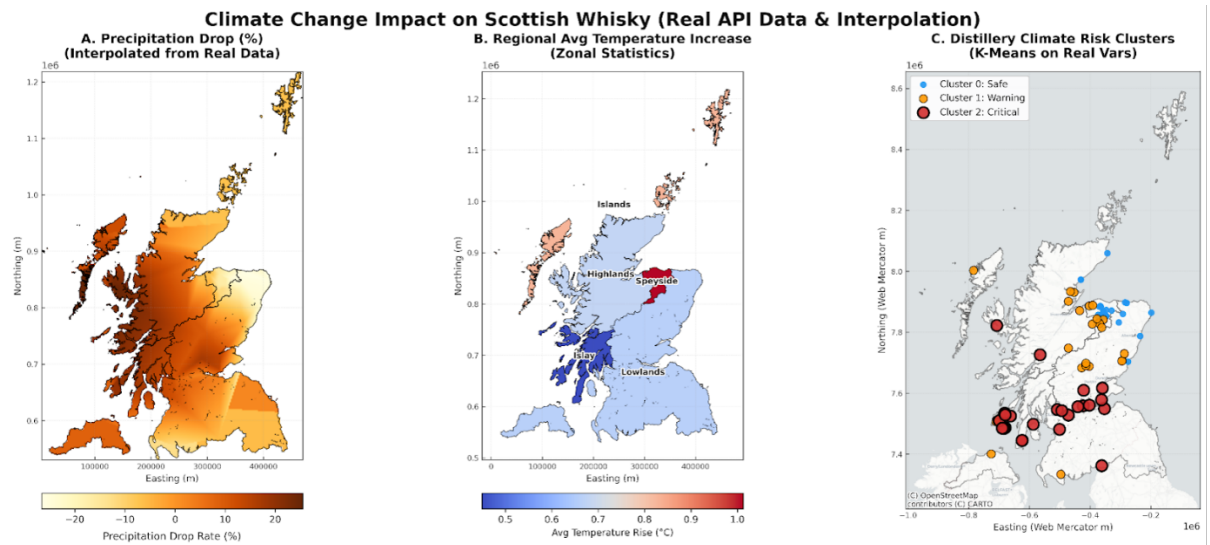


Figure 4: Spatial interpolation of baseline observed climate variables paired with tri-level K-means distillery risk clustering.

Spatial interpolation of baseline climate data (Figure 4) confirms distinct geographic divisions within the industrial landscape. Panel A and Panel B demonstrate that the axis of environmental exposure separates into a coastal precipitation drop concentrated in the south-west and a distinct interior temperature elevation ($\sim 1.0^{\circ}\text{C}$) centered squarely over the Speyside basin. Crucially, the abrupt linear discontinuities visible along the lower-left and north-western coastal bounds in Panel A are identified as edge-effect extrapolation artifacts of the IDW model, rather than physical climate phenomena.

3.2 CMIP6 Future Climate Projections (2050 Horizon)

Integrating the 2050 CMIP6 trajectory model reveals significant shifts in the regional environmental baselines.

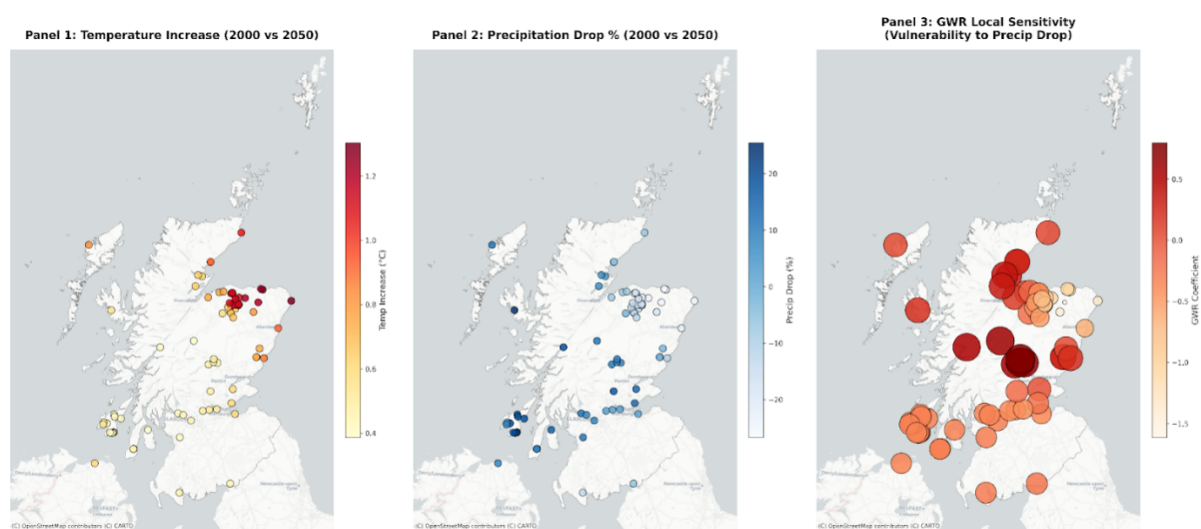


Figure 5: CMIP6-based regional future climate projections for the 2050 horizon mapping localized parameters.

As presented in Figure 5, Panel 1 exhibits a pronounced temperature expansion of up to 2.6°C localized over the northern islands and the Outer Hebrides. Simultaneously, Panel 2 indicates a structural precipitation decline approaching 40% trending toward the south-eastern Lowlands. Panel 3 highlights a major clustering of large GWR coefficients moving along the north-eastern Highland boundary. These values represent scenario tendencies and point estimates derived from the MPI-ESM1-2-XR trajectory.

3.3 Continuous Surface Modeling of the Composite Index (CVI)

Compiling the separate environmental grids yields the continuous 0 to 100 CVI surface, mapping total projected exposure across the Scottish landscape.

Quantitative Climate Vulnerability Index (CVI) Across Scotland

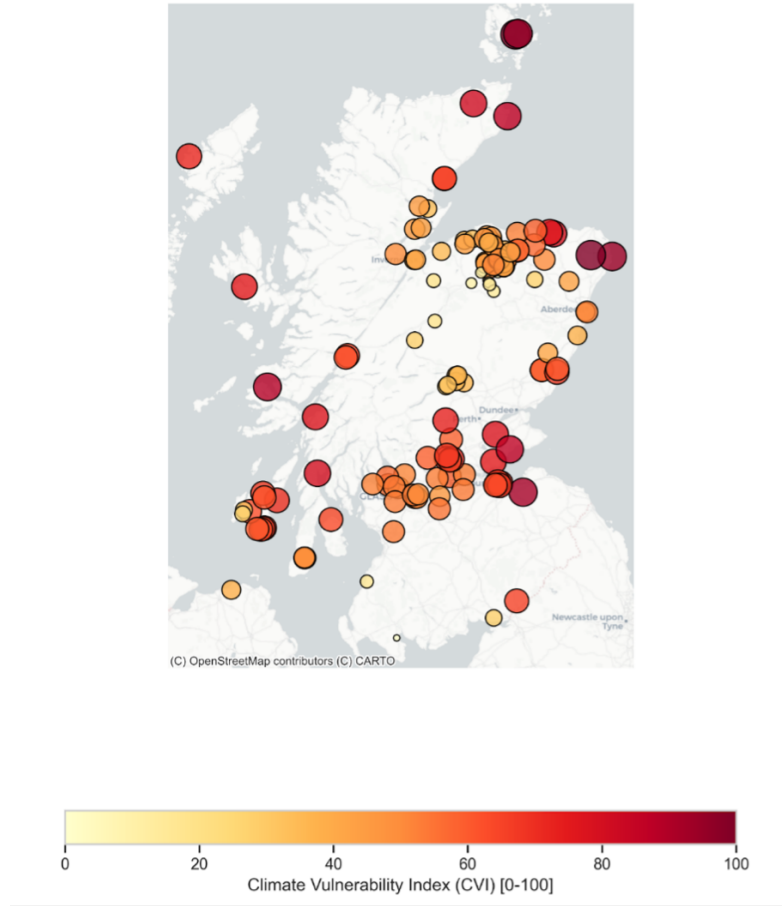


Figure 6: Continuous surface visualization of the finalized Composite Climate Vulnerability Index (CVI) across Scotland.

As shown in Figure 6, high-exposure zones falling within the 80 to 100 band are located on the far-northern Orkney margin, the western Islay region, and localized pockets of the southern Lowlands. Conversely, the north-eastern Speyside basin is situated within a stable, low-exposure band ranging from 30 to 50. However, the highest-exposure designations overlap with coastal boundaries prone to the IDW extrapolation artifacts discussed in Section 3.2. Consequently, these extreme coastal edge estimates carry wider confidence intervals, whereas the low-exposure classification for interior Speyside remains methodologically robust.

4 Discussion

4.1 Variance Decomposition and Index Structure

To understand the underlying drivers of the index, we perform a comprehensive variance decomposition. The index is cross-tabulated against its constituent components, capital scales, and sensory flavour profiles.

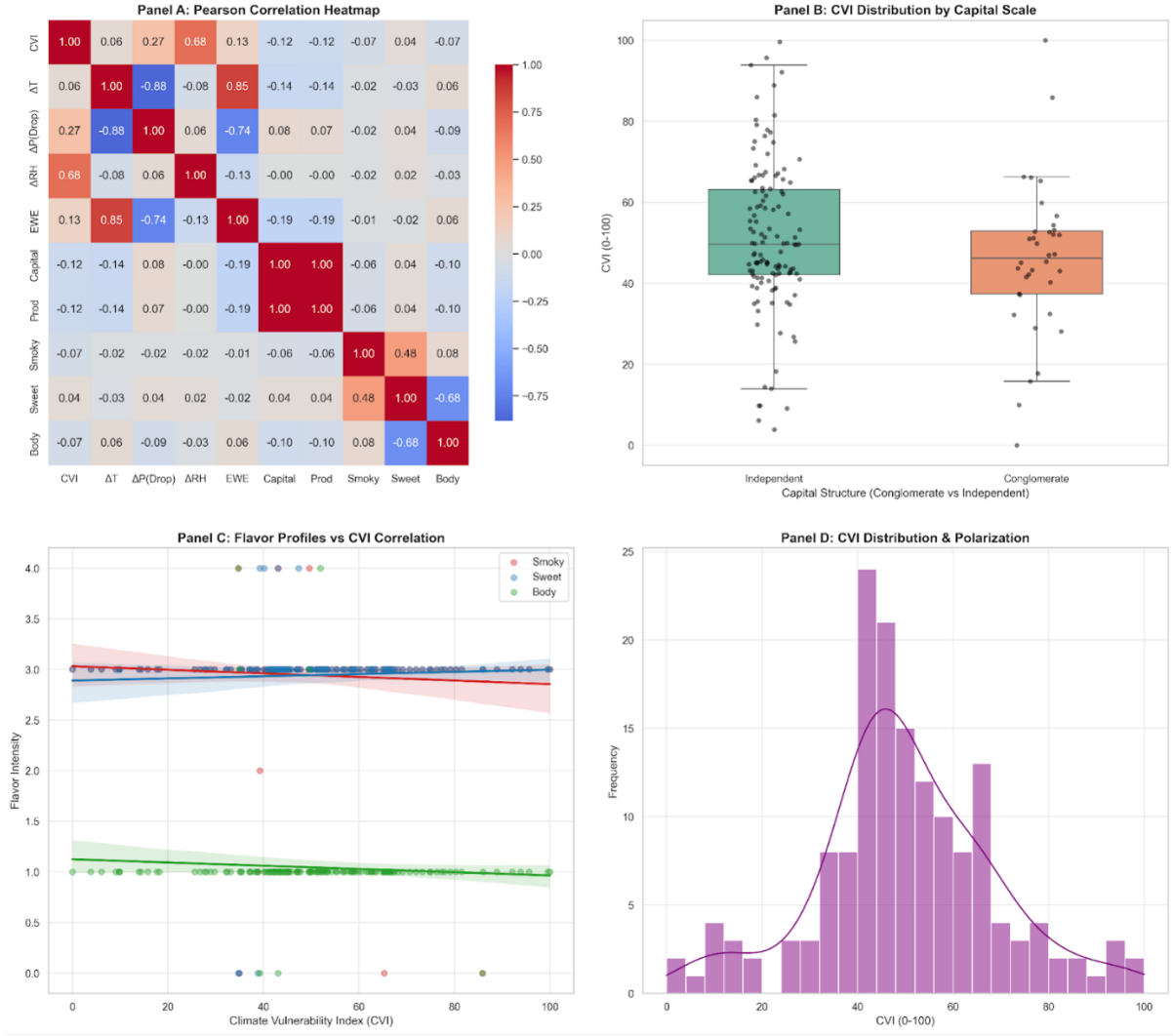


Figure 7: Multivariate diagnostic panel: Pearson correlation heatmap (Panel A), CVI distribution by capital scale (Panel B), flavour profile correlation models (Panel C), and CVI frequency distribution (Panel D).

As detailed in the Pearson correlation heatmap (Figure 7, Panel A), the finalized CVI correlates strongly with relative-humidity modifications (ΔRH , $r = 0.68$), moderately with precipitation decline (ΔP , $r = 0.27$), and weakly with temperature rise (ΔT , $r = 0.06$). Because the CVI is inherently a weighted function of these identical inputs, these metrics do not represent independent external validation; rather, they serve as an internal variance decomposition indicating which sub-component structurally dominates index variance.

This reveals a notable divergence between the initial design weights and empirical behavior. Although precipitation decline (ΔP) was assigned the highest nominal weight (0.40), relative humidity (ΔRH , $r = 0.68$) exerts the dominant statistical influence on final CVI variation. This behavior occurs because the variables were aggregated in their raw units without prior standardization, allowing the higher-variance relative humidity data to drive the index regardless of nominal weighting. Furthermore, the strong negative correlation between ΔT and ΔP ($r = -0.88$) indicates high multicollinearity, confirming that the deployment of GWR is justified solely by spatial non-stationarity rather than independent predictor behavior.

4.2 Capital Structure and Geographic Confounding

Panel B of Figure 7 illustrates that conglomerate-owned distillation facilities cluster inside a tight, mid-exposure band (40 to 50), whereas independent distilleries exhibit high dispersion extending into extreme exposure bands (up to 100). However, the direct global correlation between capital scale and the CVI is exceptionally weak ($r = -0.12$).

Thus, the intuitive narrative that large corporate conglomerates actively purchase or construct safer geographic infrastructure is not supported by the empirical data. A more conservative explanation lies in geographic confounding: independent distilleries are historically more common on peripheral islands and isolated coastal positions (geographically exposed zones, e.g., Islay), while corporate conglomerates exhibit high concentration within the interior Speyside basin (climatically low-exposure zones). This aligns with viticultural observations by van Leeuwen and Seguin (2006) noting that elite terroir frequently co-occurs with historically and socio-economically favorable geographic routing, representing structural co-occurrence rather than direct linear causation [2].

4.3 Flavour Profiles and Policy Implementations

As modeled in Panel C, the intensity vectors for smoky, sweet, and body sensory characteristics remain fundamentally flat across the CVI continuum. This flat correlation indicates that there is no linear relationship between current flavour profiles and future climate change exposure. These data do not support the hypothesis that flavour typologies will degrade gradually or disrupt abruptly at a specific structural threshold. Nonetheless, physical warehouse modifications noted by Adaptation Scotland (2025)—wherein altered humidity balances change evaporation ratios between alcohol and water within the cask—provide qualitative grounds for potential future flavour shifts [1].

In terms of distribution, Panel D reveals that the CVI distribution is close to bimodal, displaying a primary peak near the median and smaller sub-peaks at both tails. This indicates that the Scottish whisky industry is experiencing an environmental polarization into low-exposure and high-exposure clusters, mirroring regional disparities observed by LaMunyon (2025) [3].

From a policy perspective, these high-resolution spatial outputs can assist institutional bodies in prioritizing the targeted adaptation strategies outlined by Adaptation Scotland (2025). These include expanding rainwater harvesting infrastructures, developing drought-resistant barley strains via the International Barley Hub, diversifying production with alternative grains, and executing the Scotch Whisky Association’s (SWA) Water Stewardship Framework [1]. Because this index models environmental exposure rather than realized production loss, actual economic targeting should be validated against external outcome variables, including SEPA water-scarcity classifications and documented historical instances of climate-induced production stoppages.

5 Limitations

1. **Self-Reference of the Index:** The CVI is a weighted sum of internal components ($\Delta P, \Delta T, \Delta RH$). Its strong relationships with these components reflect internal index structure rather than independent validation, meaning it cannot be used to isolate external causal drivers.

2. **Absence of an External Outcome Variable:** Lacking empirical validation against actual production metrics, output reductions, or historical closures, it remains untested whether high exposure directly causes operational failure. Thus, 'exposure' or 'hazard' represents a more accurate scientific label than 'vulnerability'.
3. **Boundary Interpolation Uncertainty:** The peripheral island and coastal zones flagged with high CVI values overlap significantly with the spatial zones prone to IDW extrapolation artifacts. Consequently, the edge values carry wider confidence intervals than interior regions.
4. **Weighting and Standardization Mismatch:** Because variables were aggregated in raw mathematical units without standardization, the nominal design weight ($\Delta P = 0.40$) diverges from the empirical contribution (ΔRH dominant). Future iterations require full z-score standardization prior to summation.
5. **Single Model and Scenario Constraints:** The climate projection relies entirely on a single CMIP6 trajectory (MPI-ESM1-2-XR). The absolute figures (2.6°C rise, 40% precipitation decline) should be treated as point estimates; multi-model ensemble analysis remains a target for future research.
6. **Capital and Location Confounding:** Capital scale and geographic exposure co-occur due to historical factory placement. No direct causal effect of capital safeguarding operations can be inferred.
7. **Limits of the Terroir Analogy:** Viticultural terroir models assume that crops are grown in place. Because malting barley for whisky production is frequently sourced from non-local supply chains, the viticultural framework serves primarily as a thematic motivator.
8. **Operational Selection of k :** While the global silhouette metrics peak at $k = 2$, this study enforces a local parameters choice of $k = 3$ to support an interpretation-driven, tri-level risk categorization (Safe, Warning, Critical).

References

- [1] Adaptation Scotland (2025). *Industry Insights: Climate Change Risks and Adaptation in the Scotch Whisky Industry*. Edinburgh: Adaptation Scotland (Sniffer), in collaboration with the Scotch Whisky Association. Available at: <https://adaptation.scot>.
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- [3] LaMunyon, C. (2025). *Impact of Climate Change on Whisky Production* (Master of Science final project, GIST). University of Arizona. Available at: <http://hdl.handle.net/10150/677091>.